

Cytokines

- a. List and describe the general features of cytokines.
- b. Compare and contrast the role of helper T cells and cytotoxic T cells after antigen presentation.
- e. Describe assays for cytokines.
- f. Differentiate among and define the terms autocrine effect, paracrine effect, and endocrine effect.
- g. Describe and compare the different classes of cytokines based on their function.
- h. Describe the function of cytokines involved in innate immunity.
- i. Describe the function of cytokines involved in adaptive immunity.

cytokines activate B cells

Nomenclature

- **Monokines:** Ck produced by mononuclear cells
- **Lymphokines:** CK produced by lymphocytes
- **Cytokines** including interleukins

General Properties

Structure: Polypeptide produced in response to microbes

Stability and storage

- Secretion is brief → ends itself
- Short half life, Self-limited event
- Not stored as preformed mediators (no storage)
- Unstable mRNA, activation is transient
↓
breaks down

cytokines are not lipids nor carbohydrates they're all proteins (polypeptides)

Action

cytokine

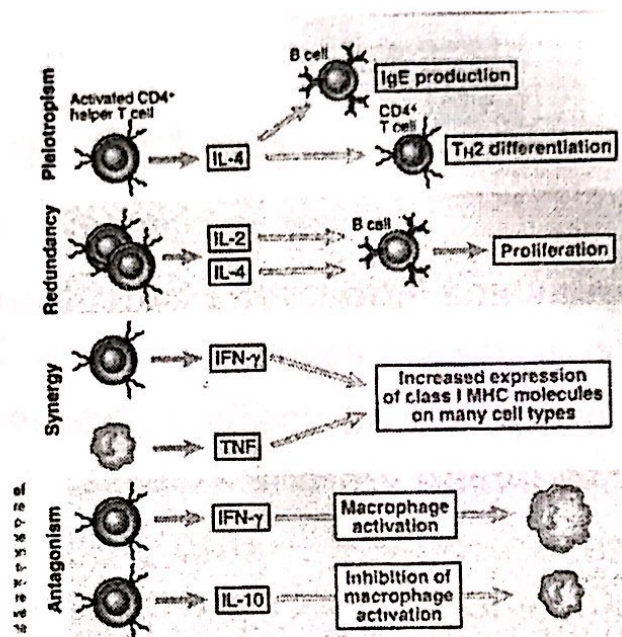
a. **Pleiotropic:** ability of one CK to act on different cell types

IL-4 $\xrightarrow{\text{acts on}}$ B cells to secrete IgE
 $\xrightarrow{\text{acts on}}$ T cells → Th2

IL-2 affect osteoblasts and T cell differentiation

b. **Redundancy:** multiple CK have common function

IL-2, IL-4 both affect B cell differentiation



Action

One CK influence the action of the other

- additive effect:** IL-2 and IL-4 both $\rightarrow$ if IL-2 has 20% effect and IL-4 40%, the effect is 60%.
- Synergistic effect:** IL-1 upregulates IL-2 and IL-2R $\rightarrow$ if IL-1 has 10% effect and IL-2 has 40%, the total effect is 80%, because IL-1 activates IL-2.
- Antagonistic effect:** IL-10 inhibit Mac and IFN- γ activate Mac

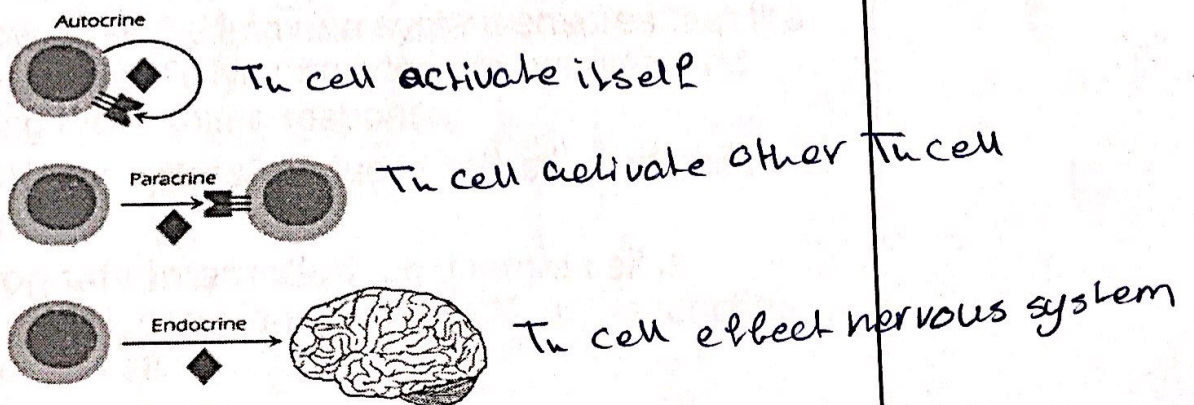
Site of Action

a. **Locally** that comprise

autocrine effect: one T cell produce IL-2 which in turn activates the same cell

paracrine effect: one cell produce IL-2 which in turn activates another neighboring T cell.

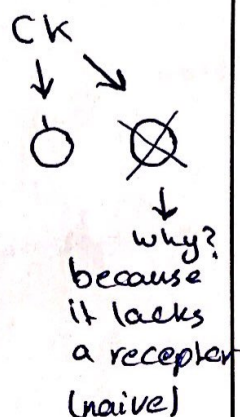
b. **systemic effect: Endocrine** function at high concentration as TNF has local and systemic effect



cytokines are

LIKE HORMONES:

- act at very low conc. 10^{-15} ^{molar} M more potent than enzymes, receptors and antibodies
- bind specific receptors: this explain why only cells with specific R are activated
- Endocrine function



Not specific:

But how does the immune system ensures that the Ag-nonspecific lymphocytes are not activated during the immune response:

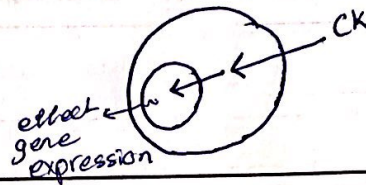
- CK Receptors** produced by activated cells only
- cognate interaction:** only target cell is affected due to high conc. of CK at the junction between cells
- short half life**

what determines the specificity are receptors

It are not specific

7. Mechanism of action: درجه

- Establishment of conc. Gradient in cascade fashion during the immune response.
- Makes changes in gene expression



CK facilitate Innate I.R.

associated with inflammation

group 1: *IL-1, IL-6 and TNF- α : produced by phagocytes

a. fever as endogenous pyrogenes (fever)

Overexposure to UV (sunburn), keratinocytes damage—release of

← IL-1 causing chills and malaise dendritic cells

b. acute phase proteins: produced by hepatocytes (liver)

C-reactive protein (CRP) bind phosphorycholine on bac
Mannan-binding lectin (MBL): opsonin and Complement activation

c. increase in phagocytes

d. Proinflammatory CK: ↑ vascular permeability, Expression of adhesion molecules on EC, activates phagocytes

group 2: *IFN- α and β : produced by many cell types after viral infection to protect the uninfected cells (MHC class I, NK, CD8 T)

group 3: *IL-17: stimulate the mobilization and de novo generation of neutrophils by G-CSF participate in inflammation and auto immune disease

group 4: *Chemokines

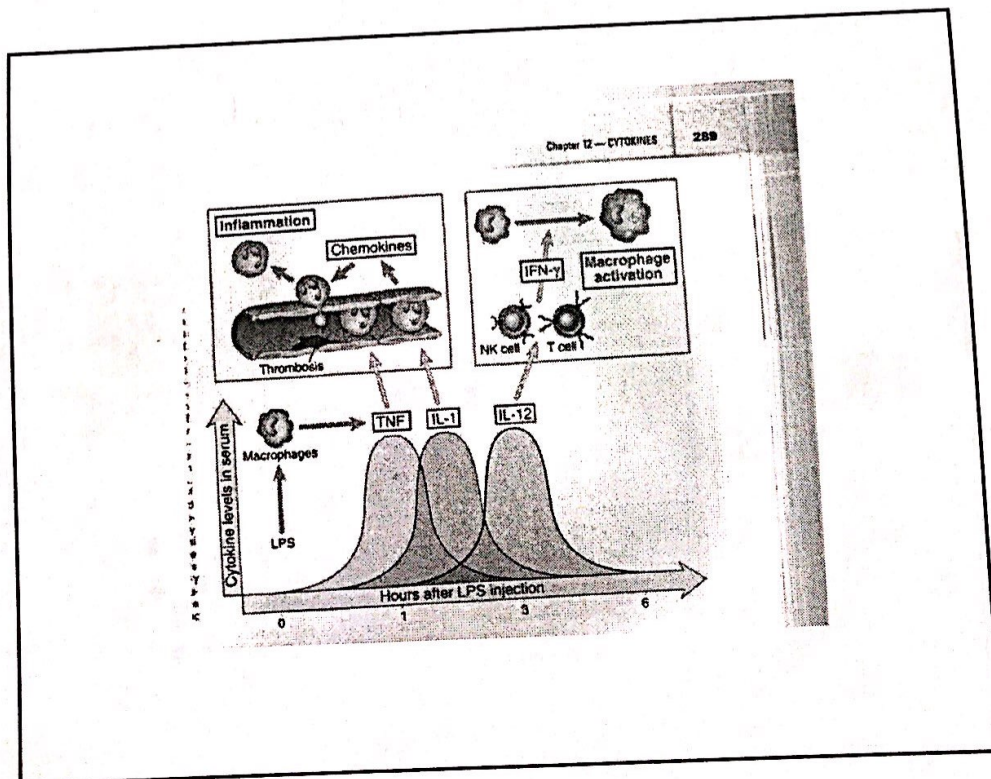
effects the brain
the most famous acute protein

→ when there's a bacteria outside the cell we contents of the cell need to pass through the EC, IL make spaces in the EC allowing cell contents to pass

Table 12-3. Cytokines of Innate Immunity

Cytokine	Size (kD)	Principal cell source	Principal cellular targets and biologic effects
Tumor necrosis factor (TNF)	17 kD; 51 kD homotrimer	Macrophages, T cells	Endothelial cells: activation (initiates coagulation) Neutrophils: activation Hypothalamus: fever Liver: synthesis of acute phase proteins Muscle, fat: catabolism (cachexia) Many cell types: apoptosis
Interleukin-1 (IL-1)	17 kD mature form; 33 kD precursors	Macrophages, endothelial cells, some epithelial cells	Endothelial cells: activation (initiates coagulation) Hypothalamus: fever Liver: synthesis of acute phase proteins
Chemokines (see Table 12-4)	8–12 kD	Macrophages, endothelial cells, T cells, fibroblasts, platelets	Leukocytes: chemotaxis, activates migration into tissues
Interleukin-12 (IL-12)	Heterodimer of 35 kD + 40 kD subunits	Macrophages, dendritic cells	T cells: Th1 differentiation NK cells and T cells: IFN- γ synthesis Increased cytotoxic activity
Type I IFNs (IFN- α , IFN- β)	IFN- α : 15–21 kD IFN- β : 20–25 kD	IFN- α : macrophages IFN- β : fibroblasts	All cells: antiviral state, increase MHC expression NK cells: activation
Interleukin-10 (IL-10)	Homodimer of 34–40 kD; 18 kD subunits	Macrophages, T cells (mainly regulatory T cells)	Macrophages, dendritic cells: suppress IL-12 production and expression of costimulators and class II MHC
Interleukin-6 (IL-6)	19–26 kD	Macrophages, endothelial cells, T cells	Liver: synthesis of acute phase proteins B cells: proliferation of antibody-producing cells
Interleukin-15 (IL-15)	13 kD	Macrophages, others	NK cells: proliferation T cells: proliferation (memory CD8)
Interleukin-18 (IL-18)	17 kD	Macrophages	NK cells and T cells: IFN- γ synthesis
Interleukin-23 (IL-23)	Heterodimer of unique 19 kD subunit and 40 kD subunit of IL-12	Macrophages and dendritic cells	T cells: maintenance of IL-17-producing T cells
Interleukin-27 (IL-27)	Heterodimer of 28 kD and 13 kD subunits	Macrophages and dendritic cells	T cells: inhibition of Th1 cells; role in Th1 differentiation? NK cells: IFN- γ synthesis

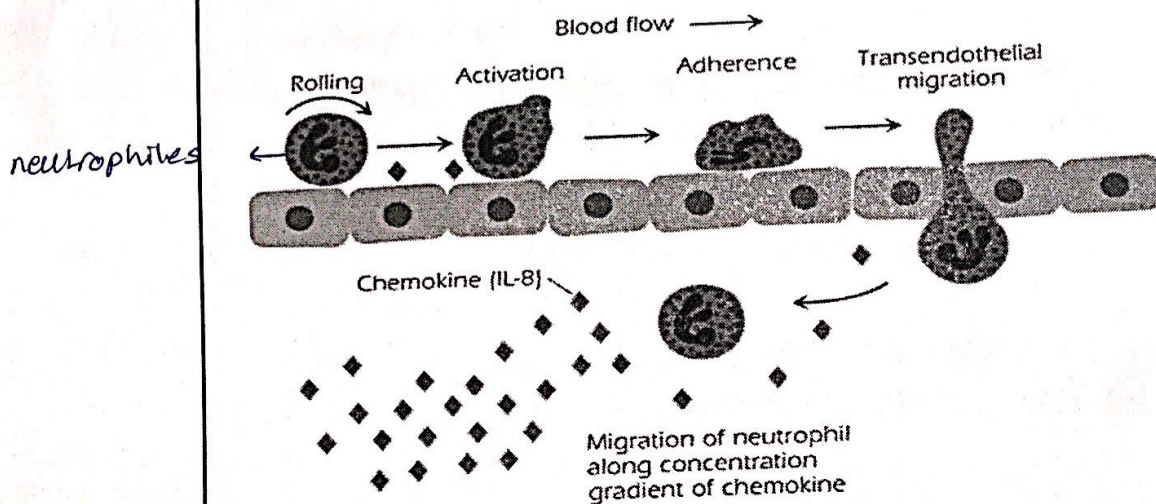
Abbreviations: IFN, interferon; kD, kilodalton; MHC, major histocompatibility complex; NK, natural killer.

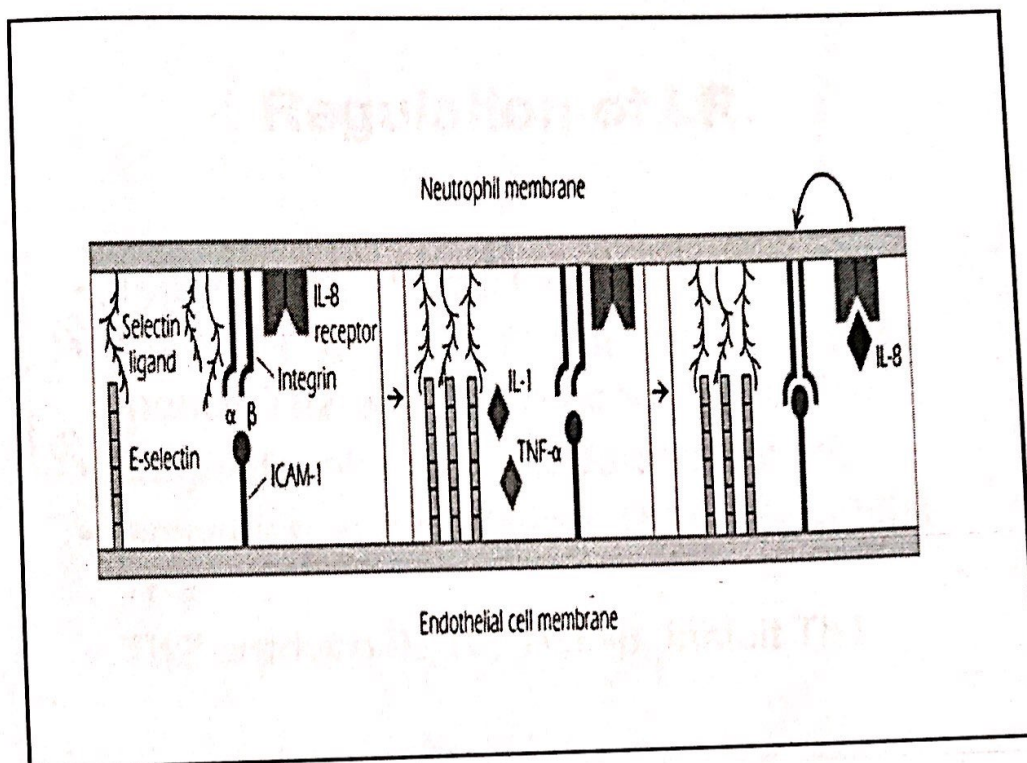


Chemokines

- Family of structurally related CK
- Low MW proteins
- Conserved sequences CC, CXC (C: Cys) $\rightarrow$ rich in Cys
- >50 are known
- Functions: chemotactic effect, expression of adhesion molecules, WBC traffic
- IL-8: produced by phagocytes, naïve T and EC
Attraction and activation of neutrophils
- RANTES: produced by naïve T, EC
Attraction of phagocytes, NK, T, activates T cells
- MIP: Mac inflammatory protein: produced by phagocytes, and EC
Attraction of phagocytes, NK
Compete with HIV to bind CCR5:
Mac-CCR5 ——— HIV or RANTES, MIP competition

there are chemokines receptors for all kinds of Chemokines





CK stimulate Hematopoiesis

Formation of blood cells

- monocytes* *granulocytes*

↑ ↑

• **M-CSF, G-CSF, MG-CSF:** support the growth and proliferation of Mono, Granulo or both, *they are now used as drugs because they activate the immune system*
 - **IL-3:** stimulate many cells Lym, RBC, Granulo
 - **IL-7:** Bone marrow stroma and thymic stroma: induce differentiation of lymphoid stem cells into T cells.
- deficiency of IL-7 means deficiency of T-cells*

Regulation of I.R.

regulation of I.R.
depend on

- Types of Ag affect Th1 or Th2
- ① • Conc. Of Ag affect Th1 or Th2: Low conc. Induce Th2 while high conc. Induce Th1
- ② • Regulate intensity and duration of I.R.
- Activation and inhibition: Th1 IFN- γ inhibit Th2
- Th2 produce IL-10, TGF- β inhibit Th1

Cytokine receptors

- Characteristics:
consist of one or more transmembrane proteins: extracellular portion binds cytokine,
cytoplasmic portion responsible for initiating intracellular signaling pathways

Cytokines and disease

- **Toxic shock due to bac exotoxins:**

- enterotoxins of m.o. (S aureus, S pyogens)
- SuperAg (Not processed, intact molecule-MHC II outside the groove)
- Large No of lymphocytes (10% of CD4+)
- High level of CK (IL-1 and TNF- α)
- Fever, diahrea, hypotension and shock

- **Septic shock due to gram negative E.coli, Endotoxins of K pneumonia, P aeruginosa, Neisseria meningitidis**

↓ Mac → K. pneumoniae
IL-1 and TNF- α

Therapy

- **Autoimmune diseases:** TNF inhibitors (Monoclonal antibody or TNF Receptors) to treat:
Rheumatoid arthritis, IFN for Multiple. sclerosis
- **Immunodeficiency diseases:** G-CSF to treat
gammaglobulenemia
- **Cancer:** Human HTLV retrovirus cause lymphoma by autocrine excessive production of IL-2 and IL-2 R that leads to uncontrolled growth
- **Transplantation:** Anti-CD25
- **Allergy:** Mab against IL-4 and IgE
- **Cellular deficiencies** arising from chemotherapy and radiotherapy: G-CSF, GM-CSF